

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Reads graph and uses $10^{\log P}$ or $10^{\log V}$ to get either P or V	AO3.4	M1	$\log_{10} P = 2.18$ $P = 151$ $\log_{10} V = -0.15$ $V = 0.708$
	Correctly obtains P and V AWRT 150 and AWRT 0.71	AO1.1b	A1	
(b)	Calculates value of gradient to find d Condone use of $\log d =$ gradient Or uses a value of c plus a P/V pair to find d	AO3.4	M1	$\log_{10} P = \log_{10} c + d \log_{10} V$ Gradient = $d = -1.4$ Intercept = $\log_{10} c$ $c = 93.3$
	Obtains correct value for d AWRT -1.4 Not necessarily a decimal	AO1.1b	A1	
	Calculates value of intercept to find $\log_{10} c$ Or uses a value of d plus a P/V pair to find c	AO3.4	M1	
	Calculates correct value for c AWRT 93	AO1.1b	A1	
(c)	Uses their values of c and d in the formula $P = cV^d$	AO1.1a	M1	$P = 93.3 \times 2^{-1.4}$ $= 35.4$ kilopascals
	Obtains P value, including units AWFW 30 to 40	AO3.2a	A1	
				Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model to form one correct equation (PI by $a =$	AO3.1b	M1	$290 = a + b$ $331 = ae^{0.1} + be^{0.2}$

	200) (ACF) Accept $1.105a + 1.221b = 331$			
	Forms a second correct equation (ACF)	AO1.1a	M1	$290e^{0.1} = ae^{0.1} + be^{0.1}$ $b = \frac{(331 - 290e^{0.1})}{(e^{0.2} - e^{0.1})} = 90.3$ $= 90$ to the nearest integer so $a = 200$
	Obtains correct a AWRT 200 and b AWRT 90 (AG) Only award if both previous M1's achieved. Do not award marks retrospectively for correct values of a and b used in part (b)	AO1.1b	A1	
(b)	Substitutes $t = 3$ and evaluates CAO	AO3.4	B1	
(c)	Forms inequality (accept $<$ or $>$) (condone use of equation) FT 'their' value of a , but b must be 90	AO1.1a	M1	$90e^{0.2t} > 200e^{0.1t}$ $e^{0.1t} > \frac{200}{90}$ $0.1t > \ln\left(\frac{200}{90}\right)$ $t > 10\ln\left(\frac{200}{90}\right) = 7.985$ Just less than 8 so during 2024
	Uses logs or calculator to solve 'their' inequality (or equation) If using trial and error must see $t = 7$ and $t = 8$ tested	AO1.1a	M1	
	Interprets final result. (Do not accept 2025)	AO3.2a	A1	
(d)	Gives one limitation of the model. Eg. Model must break down as both n_A and n_B will tend to infinity / model assumes nothing changes / no attempt to control the diseases / all the trees have died / finite number of trees / cure for the disease might be found / other factors such as drought could affect the model / etc.	AO3.5b	E1	Eventually all of the trees will die so the model will no longer be accurate.
Total 8 marks				

Q3.

Marking Instructions	AO	Marks	Typical Solution
Takes logs of an equation. Must be correct use of logs.	AO1.1a	M1	$y = e^{x-4}$ $\ln y = x - 4$ $4 + \ln y = x$ $f^{-1}(x) = 4 + \ln x, x > 0$
Obtains correct inverse function in any correct form	AO1.1b	A1	
Deduces correct domain	AO2.2a	B1	
			Total 3 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates 2^t or 2^{-t} to obtain $\pm A \ln 2 \times 2^{\pm t}$	AO1.1a	M1	$\frac{dy}{dt} = (3 \ln 2) 2^t$ $\frac{dx}{dt} = (-4 \ln 2) 2^{-t}$ $\frac{dy}{dx} = \frac{(3 \ln 2) 2^t}{(-4 \ln 2) 2^{-t}}$ $= -\frac{3}{4} \times 2^{2t}$
	Obtains $\frac{dy}{dt} = (\pm A \ln 2) 2^t$ and $\frac{dx}{dt} = (\pm B \ln 2) 2^{-t}$	AO1.1b	A1	
	Uses chain rule with correct $\frac{dy}{dt}$ and $\frac{dx}{dt}$ and completes rigorous argument to obtain fully correct printed answer	AO2.1	R1	
(b)	Rearranges to write 2^{-t} in terms of x or 2^t in terms of y Or Writes given expression in terms of t	AO3.1a	M1	$2^t = \frac{y+5}{3}$ $2^{-t} = \frac{x-3}{4}$ $1 = \left(\frac{y+5}{3} \right) \left(\frac{x-3}{4} \right)$ $12 = xy + 5x - 3y - 15$ $xy + 5x - 3y = 27$ ALT $xy + ax + by = (4 \times 2^{-t} + 3)(3 \times 2^t - 5) + a(4 \times 2^{-t} + 3) + b(3 \times 2^t - 5)$ $= 12 - 15 + (4a - 20)2^{-t} + (3b + 9)2^t + 3a - 5b$
	Eliminates t Or Compares coefficients PI by $a = 5$ or $b = -3$	AO1.1a	M1	
	Completes rigorous argument to obtain correct values of a , b and c and write the Cartesian equation in the required form ISW	AO2.1	R1	

			$a = 5, b = -3$ $xy + 5x - 3y = -3 + 15 + 15$ $= 27$
Total 6 marks			

Q5.

Marking Instructions	AO	Marks	Typical Solution
States the correct gradient of the curve	AO1.2	B1	Grad of curve = $2e^{2x}$
Forms an equation using 'their' gradient of the curve and puts it equal to $\frac{1}{2}$	AO1.1a	M1	= grad of tangent so $2e^{2x} = \frac{1}{2}$
Takes a log of each side of 'their' equation and uses law of logs to obtain equation in x	AO1.1a	M1	$e^{2x} = \frac{1}{4} \Rightarrow 2x = \ln\left(\frac{1}{4}\right)$
Obtains a correct exact value for x	AO1.1b	A1	$\Rightarrow x = \frac{1}{2} \ln\left(\frac{1}{4}\right) = \ln\left(\frac{1}{2}\right) = -\ln 2$
Substitutes 'their' value of x and obtains y value and hence the coordinates (follow through provided values are exact)	AO1.1b	A1F	$y = e^{2x} = \frac{1}{4}$ $\left(-\ln 2, \frac{1}{4}\right)$
			Total 5 marks

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct value CAO	AO3.4	B1	50
(i)				
(ii)	States correct integer value CAO	AO3.4	B1	609
(b)	Forms correct equation and rearranges to obtain $e^{0.5t} = \dots$	AO3.4	M1	$150 = 50e^{0.5t}$ so $e^{0.5t} = 3$

	Obtains the correct solution. Must give answer to 3 sf	AO1.1b	A1	$t = 2\ln 3 = 2.20$
(c)	1 mark for any clear valid reason, must be set in context of the question	AO3.5b	E1	No constraint on the number of rabbits (ie could go off to infinity) OR Model is only based on the 3 years of the study. Things may change OR Continuous model but number of rabbits is discrete OR Ignores extraneous factors such as disease, predation, limited food supply
(d)	Forms an equation with exponentials by letting $R = C$ PI	AO3.4	M1	$1000e^{0.1t} = 50e^{0.5t}$ $20 = e^{0.4t}$
	Solves 'their' equation correctly	AO1.1a	M1	$t = \ln 20 \div 0.4$ $= 7.49$
	States correct answer as the year 2023 CAO NMS scores full marks for 2023	AO3.2a	A1	2023
Total 8 marks				

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Makes a deduction about the lower bound of the function (4)	AO2.2a	B1	The range of f is the set $\{y \in \mathbb{R} : y > 4\}$
	Correctly states the range of f using set notation	AO2.5	B1	
(b)	States correctly the set they gave in part (a)	AO1.2	B1F	$\{y \in \mathbb{R} : y > 4\}$
(i)				
(ii)	Interchanges x and y at	AO1.1a	M1	$y = 4 + 3^{-x}$

	any stage			$x = 4 + 3^{-y}$
	Rearranges and takes logs	AO1.1a	M1	$3^{-y} = x - 4$ $-y = \log_3(x - 4)$
	Obtains correct expression from completely correct working for $f^{-1}(x)$, notation correct throughout	AO1.1b	A1	$f^{-1}(x) = -\log_3(x - 4)$
(c)	Obtains $gf(x)$	AO1.1b	B1	$gf(x) = g(4 + 3^{-x})$ $= 5 - (4 + 3^{-x})^{0.5}$
(i)	Forms equation and rearranges using 'their' $gf(x) = 2$	AO1.1a	M1	$5 - (4 + 3^{-x})^{0.5} = 2$ $(4 + 3^{-x}) = 9$
(ii)	Correctly rearranges to get a single exponential term where logs can be taken. (Follow through provided 'their' equation requires the use of logs.)	AO1.1b	A1F	$3^{-x} = 5$ $x = -\log_3 5$
	Obtains correct solution	AO1.1b	A1	
Total 10 marks				

Q8.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds a difference between 2 terms	AO3.1a	M1	$3e^p - 5 = 5 - 3e^{-p}$ (*)
	Forms an equation using two differences	AO3.1a	M1	$3e^p - 10 + 3e^{-p} = 0$ $3e^{2p} - 10e^p + 3 = 0$
	Forms a quadratic equation in e^p	AO1.1a	M1	$e^p = \frac{1}{3}, 3$
	Obtains a correct quadratic equation	AO1.1b	A1	$p = \ln \frac{1}{3}, \ln 3$
	Obtains 2 correct solutions for e^p from 'their' quadratic FT only applies if previous mark has been awarded	AO1.1b	A1F	ALT to (*) $2(5 - 3e^{-p}) = 3e^p - 3e^{-p}$ Or $2(3e^{-p} - 5) = 3e^p - 3e^{-p}$
	Obtains final answers in an exact form FT applies if previous mark	AO2.2a	A1F	

	has been awarded			
(b)	Finds a ratio between two consecutive terms (no requirement to use a and r)	AO3.1a	M1	Assume it is possible that $3e^{-q}$, 5 and $3e^q$ are three consecutive terms of a geometric sequence
	Compares two ratios (could be ratios of successive terms, no requirement to use a and r)	AO3.1a	M1	$a = 3e^{-q}$, $ar = 5$, $ar^2 = 3e^q$ $\frac{ar}{a} = \frac{5}{3e^{-q}} \Rightarrow r = \frac{5e^q}{3}$ $\frac{ar^2}{ar} = \frac{3e^q}{5} \Rightarrow r = \frac{3e^q}{5}$ $\frac{5}{3e^{-q}} = \frac{3e^q}{5} \Rightarrow 25 = 9$
	Identifies a contradiction	AO2.1	R1	
	Draws a conclusion about k	AO2.4	R1	This is a contradiction therefore $3e^{-q}$, 5 and $3e^q$ cannot form three consecutive terms of a geometric sequence.
Total 10 marks				

Q9.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Integrates both components with at least one correct	AO1.1a	M1	$\mathbf{r} = \int 40e^{-0.2t} dt \mathbf{i} + \int 50(e^{-0.2t} - 1) dt \mathbf{j}$ $= (-200e^{-0.2t} - c) \mathbf{i} + (-250e^{-0.2t} - 50t + d) \mathbf{j}$
	Obtains correct terms. (condone missing constants)	AO1.1b	A1	$t = 0, \mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow c = 200, d = 250$ OR
	Evaluates both constants (or uses definite integration) using 'their' expression for $\mathbf{r}$	AO3.4	M1	$\mathbf{r} = \int_0^t 40e^{-0.2t} dt \mathbf{i} + \int_0^t 50(e^{-0.2t} - 1) dt \mathbf{j}$ $= [-200e^{-0.2t} \mathbf{i} - (-250e^{-0.2t} - 50t) \mathbf{j}]_0^t$
	Obtains correct expression	AO1.1b	A1	$\mathbf{r} = 200(1 - e^{-0.2t}) \mathbf{i}$ $+ (250 - 250e^{-0.2t} - 50t) \mathbf{j}$
(b)	Forms equation to find t based on horizontal component	AO3.4	M1	$200(1 - e^{-0.2t}) = 100$ $t = 5 \ln 2 = 3.4657$ $y = 250 - 250e^{-0.2 \times 5 \ln 2} - 50 \times 5 \ln 2$ $= -48.3$
	Obtains correct time	AO1.1b	A1	
	Substitutes 'their' time into vertical component	AO1.1a	M1	

	Obtains correct displacement for 'their' time (to 1 sf, must have metres) FT only if both M1 marks have been awarded	AO3.2a	A1F	The parachutist has a vertical displacement of 50 m below the origin
(c)	Identifies vertical component of the velocity as first step	AO2.4	M1	$\frac{d}{dt} (50(e^{-0.2t} - 1)) = -10e^{-0.2t}$
	Differentiates vertical component of velocity correctly	AO1.1b	A1	As there is no initial vertical component of velocity (and hence no air resistance) the initial acceleration is only due to gravity. Hence g is taken as 10 m s^{-2}
	Considers implication of initial motion and reaches correct conclusion	AO2.2a	R1	
Total 11 marks				

Q10.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving $\frac{dA}{dt}$, A and a constant of proportionality.	AO3.3	M1	$\frac{dA}{dt} \propto A$ $\Rightarrow \frac{dA}{dt} = kA$ $\Rightarrow \int \frac{1}{A} dA = \int k dt$
	Separates variables	AO1.1a	M1	$\Rightarrow \ln A = kt + c$ $\Rightarrow A = e^{kt+c}$
	Integrates both of 'their' sides correctly	AO1.1b	A1F	$\Rightarrow A = Be^{kt}$ AG
	Constructs a rigorous mathematical argument that supports use of the given model. AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
(b)	States correct value of B	AO1.1b	B1	$B = 0.25$ or $B = \frac{1}{4}$
(i)				
(ii)	Uses $t = 20$ and $A = 0.5$ to	AO3.1b	M1	When $t = 20$, $A = 0.5$

	find k			$\Rightarrow 0.5 = 0.25e^{20k}$
	Finds correct value of k	AO1.1b	A1	$\Rightarrow 20k = \ln 2$
	Substitutes 'their' k to get A in terms of t	AO1.1a	M1	$\Rightarrow k = \frac{1}{20} \ln 2$
	Constructs rigorous and convincing argument to show $A = 2^{\frac{t}{20}-2}$ Using correct notation throughout. AG	AO2.1	R1	$\Rightarrow A = 2^{-2} \times 2^{\frac{t}{20}}$ $A = 2^{\frac{t}{20}-2}$ AG
(iii)	Uses the model to set up correct equation and attempt to find t	AO3.4	M1	$2\pi = 2^{\frac{t}{20}-2}$
	Finds correct value of t	AO1.1b	A1	$t = 93.03$ days

(c)	States any sensible and relevant limitation of the model that is specified in terms of the pond, area, weed, rate of change or time.	AO3.5b	E1	Model predicts that the area of weed will increase without limit and this is not possible since the area of the pond is 4π
(d)	Any sensible and relevant refinement to the model that is specified in terms of the pond, area, weed, rate of change or time	AO3.5c	E1	Introduce a limiting factor such as fish eating weed or rate of growth decreases as surface area covered
				Total 13 marks

Q11.

- (a) Stretch(**I**) in y -direction(**II**) scale factor 3(**III**)

OE Need (I) and either (II) or (III)

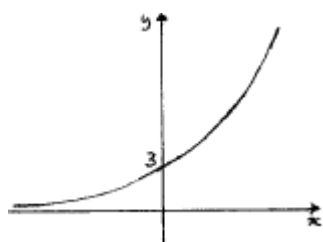
M1

*All correct. Need (I) and (II) and (III)
[>1 transformation scores 0 / 2]*

A1

2

- (b)



Shape with indication of correct asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis, and only intercept is 3 stated / indicated

B1

2

(c) $3 \times 4^x = 4^{-x}$

OE eqn. in x

M1

$$\log 3 + \log 4^x = \log 4^{-x}$$

Log Law 1 (or Law 2 applied to $\frac{4^x}{4^{-x}} = 3$ or $\frac{1}{3}$ OE)
used correctly or correct rearrangement to $4^{2x} = 1/3$
OE simplified e.g. $16^x = 3^{-1}$ or $4^x = (1/\sqrt{3})$

m1

$$\log 3 + x \log 4 = -x \log 4$$

Log Law 3 applied correctly twice (dependent on both M1 & m1) or a correct method using logs to solve an eqn. of form $a^{kx} = b$, $b > 0$ (including case $k = 1$) (dependent on M1 and valid method to a^{kx})

m1

$$x = \frac{-\log 3}{2 \log 4} \quad \left(= \frac{-\log 3}{\log 16} \right)$$

Correct expression for x or for $-x$

e.g. $x = \frac{1}{2} \log_4 \left(\frac{1}{3} \right)$

PI by correct 3sf value or better

A1

$$x = -0.396(2406 \dots) = -0.396 \text{ (to 3sf)}$$

If logs not used explicitly then max of M1m1m0.

A1

5

[9]

Q12.

(a) (i) $t = 0 \quad N = 50$

B1

1

(ii) $t = 24 \quad N = 345$
Must be 345 (not 345.2534..)

B1

1

(iii) $1 + 9e^{\frac{t}{8}} = \frac{500}{400} \Rightarrow 9e^{\frac{t}{8}} = \frac{1}{4}$
Correct algebra seen

M1

$$e^{\frac{t}{8}} = 36$$

$$\text{Or } e^{\frac{t}{8}} = \frac{1}{36}$$

m1

$$t = 8 \ln 36$$

$$\text{or } t = 16 \ln 6$$

A1

3

(b) (i) $\frac{dN}{dt} = -500 \left(1 + 9e^{\frac{t}{8}} \right)^{-2} \left(-\frac{9}{8} e^{\frac{t}{8}} \right)$
Clear attempt at chain rule or quotient rule.

M1

A1

$$= -500 \left(-\frac{1}{8} \left(\frac{500}{N} - 1 \right) \right) \left(\frac{500}{N} \right)^{-2}$$

$$\text{Use } e^{\frac{t}{8}} = \frac{1}{9} \left(\frac{500}{N} - 1 \right) \text{ to eliminate } e^{\frac{t}{8}}$$

m1

$$= \frac{N^2}{500} \left(\frac{1}{8} \left(\frac{500}{N} - 1 \right) \right)$$

$$\frac{dN}{dt} = \frac{N}{4000} (500 - N)$$

Correct algebra to AG

A1

4

Alternatives:

Alternative 1 implicit differentiation

$$e^{-\frac{t}{8}} = \frac{500 - N}{9N}$$

Correct expressions for $e^{-\frac{t}{8}}$ and attempt to use
attempt to use implicit differentiation

(M1)

$$\frac{dt}{dN} \left(-\frac{1}{8} e^{-\frac{t}{8}} \right) = -\frac{500}{9N^2}$$

Fully correct

(A1)

use $e^{-\frac{t}{8}} = \frac{1}{9} \left(\frac{500}{N} - 1 \right)$

Attempt to eliminate $e^{-\frac{t}{8}}$ using correct expression

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to get $\frac{dt}{dN} = \frac{4000}{9N^2} \times \frac{9N}{500 - N}$

$$\frac{dN}{dt} = \frac{N}{4000} (500 - N)$$

(A1)

(4)

Alternative 2 explicit differentiation

$$t = -8 \ln \left(\frac{500 - N}{9N} \right)$$

$$\frac{dt}{dN} = -8 \left(\frac{(500-N) \left(\frac{-1}{9N^2} \right) - \frac{1}{9N}}{\left(\frac{500-N}{9N} \right)} \right)$$

Correct expression for t and attempt at differentiation
with use of chain rule and product for $\ln$ derivative.

(M1)
(A1)

$$= \frac{8}{9N} \left(9 + \frac{9N}{500-N} \right)$$

Clear fractions within fractions

(m1)

$$= \frac{8}{9N} \left(\frac{4500}{500-N} \right)$$

$$\frac{dN}{dt} = \frac{N}{4000} (500-N)$$

All correct

(A1)

(4)

Or

$$t = -8(\ln(500-N) - \ln(9N))$$

$$\frac{dt}{dN} = -8 \left(\frac{-1}{500-N} - \frac{9}{9N} \right)$$

Correct expression for t and $\ln$ derivatives,
condone sign errors

(M1)
(A1)

$$= 8 \left(\frac{1}{500-N} + \frac{1}{N} \right)$$

$$= 8 \left(\frac{N+500-N}{N(500-N)} \right)$$

Common denominator to combine fractions

(m1)

$$= \frac{4000}{N(500-N)} \Rightarrow \frac{dN}{dt} = \frac{4000}{N(500-N)}$$

All correct

(A1)

(4)

Alternative 3 solve differential equation

$$\int \frac{dN}{N(500-N)} = \int \frac{dt}{4000}$$

Separate variables, and attempt to form partial fractions and integrate to ln terms = kt + C

(M1)

$$\int \frac{1}{500} \left(\frac{1}{N} + \frac{1}{500-N} \right) dN = \int \frac{dt}{4000}$$

(A1)

$$\ln N - \ln(500 - N) = \frac{500}{4000}t + C$$

$$(t = 0 \ N = 50) \quad C = \ln \left(\frac{1}{9} \right)$$

Use (50,0) to find C and obtain $e^{\frac{1}{8}t} = f(N)$

(m1)

$$\ln \left(\frac{9N}{500-N} \right) = \frac{1}{8}t \Rightarrow \frac{9N}{500-N} = e^{\frac{1}{8}t}$$

$$N \left(9 + e^{\frac{1}{8}t} \right) = 500e^{\frac{1}{8}t}$$

$$N = \frac{500e^{\frac{1}{8}t}}{9 + e^{\frac{1}{8}t}} = \frac{500}{1 + 9e^{-\frac{1}{8}t}}$$

Manipulate correctly to original given equation.

(A1)

(4)

(ii) $\frac{d}{dN}(500N - N^2) = 500 - 2N$

Differentiate and attempt to find N at max value

M1

$$500 - 2N = 0 \Rightarrow N = 250$$

Condone $\frac{d^2}{dr^2}$ for $\frac{d}{dN}$

A1

$$9e^{\frac{T}{8}} = 1$$

m1

$$e^{\frac{T}{8}} = 9$$

$$T = 8 \ln 9 \approx 17.577$$

$T = 17$ or better

CSO

Accept 17, 18, 17.5, 17.6

A1

4

Alternative, by inspection

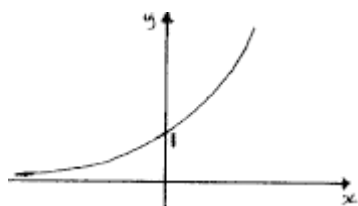
Max of $N(500 - N)$ occurs at $N = 250$

(B2)

[13]

Q13.

(a)



Correct graph, must clearly go below the intersection pt and an indication of correct behaviour of curve for large positive and large negative values of x .

Ignore any scaling on axes.

B1

Only one y -intercept, marked / stated as 1 or as coords $(0, 1)$ with graph having no other intercepts on either axes.

B1

2

(b) $9^x = 15 \Rightarrow x \log 9 = \log 15$
OE eg $x = \log_9 15$

M1

(x =) 1.23(2486...) = 1.23 to 3sf

Condone > 3sf.

Must see evidence of logs used so NMS scores 0/2

A1

2

(c) $\{f(x) =\} 9^{-x}$

OE

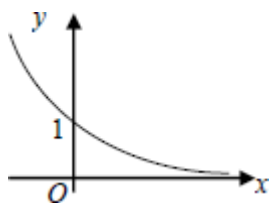
B1

1

[5]

Q14.

(a)



Correct shaped graph in 1st two quadrants only and indication of correct behaviour of curve for large positive and negative vals. of x. Ignore any scaling on axes.

B1

y-intercept indicated as 1 on diagram or stated as intercept = 1 or as coords (0, 1).

B1

2

(b) $\frac{1}{2^x} = \frac{5}{4} \Rightarrow 2^{-x} = \frac{5}{4}$ (or $2^x = \frac{4}{5}$ or $2^{2-x} = 5$)

Correct 'rearrangement' to eg $2^x = \frac{4}{5}$ or $2^{-x} = \frac{5}{4}$ or $0.5^x = 1.25$ PI or $\log 1 - \log 2^x = \log(5/4)$ or better

M1

$\log 2^{-x} = \log 1.25 \Rightarrow -x \log 2 = \log 1.25$

[$\log 2^x = \log 0.8 \Rightarrow x \log 2 = \log 0.8$]

$$[\log 2^{2-x} = \log 5 \Rightarrow (2-x) \log 2 = \log 5]$$

$$[2^x = 0.8, x \log_2 0.8]; [0.5^x = 1.25, x = \log_{0.5} 1.25]$$

Takes logs of both sides of eqn of form either $2^x = k$ or $2^{-x} = k$ OE and uses 3rd law of logs or log to base 2 (or base $\frac{1}{2}$) correctly

M1

$$x = -0.321928 \dots \text{ so } x = -0.322 \text{ (to 3sf)}$$

*Condone > 3sf [Logs must be seen to be used otherwise **max** of M1M0A0]*

A1

3

(c) $\log_a b^2 + 3 \log_a y = 3 + 2 \log_a \left(\frac{y}{a} \right)$

$$\log_a b^2 + 3 \log_a y = 3 + 2 [\log_a y - \log_a a]$$

A log law used correctly; condone missing base a .

M1

$$\log_a b^2 + 3 \log_a y = 3 - 2 \log_a a$$

A different log law used correctly condone missing base a .

M1

$$\log_a b^2 y = 3 - 2(1) \quad [\text{or } \log_a b^2 y + \log_a a^2 = 3]$$

Either a further different log law used correctly condone missing base a or $\log_a a = 1$ a stated / used.

M1

$$\Rightarrow \log_a b^2 y = 1 \quad \Rightarrow b^2 y = a$$

$\log Z = k \Rightarrow Z = a^k$ used or a correct method to eliminate logs (dep on no misapplication of any log law OE in the whole solution) Rearrangements which require only two of the above Ms to eliminate logs correctly: award the remaining M with the m mark.

M1

$$\Rightarrow y = ab^{-2}$$

ACF of RHS

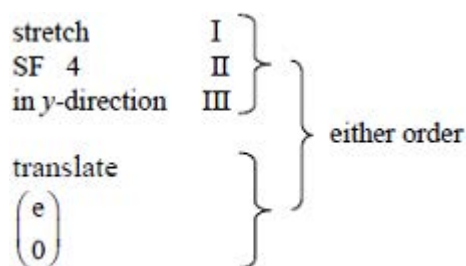
A1

5

[10]

Q15.

(a)



$$I + (II \text{ or } III)$$

M1A1

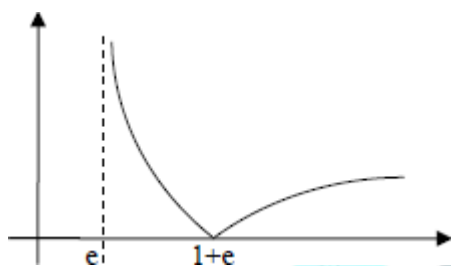
E1

accept 'e in positive x-direction'

B1

4

(b)



mod graph, in 2 connected sections, both in the first quadrant, touching x-axis

M1

curve touches x-axis at $1 + e$ (or 3.7 or better), and labelled (ignore scale)

A1

correct curvature, including at their $1 + e$, approx.
asymptote at $x = e$

A1

3

(c) (i) $|4\ln(x - e)| = 4$

$$4\ln(x - e) = 4 \quad 4\ln(x - e) = -4 \text{ or better}$$

must see 2 equations, condone omission of brackets

M1

$$(x =) 2e \quad \text{do not ISW}$$

accept values of AWRT 5.42, 5.43, 5.44

A1

$(x =) e + e^{-1}$ or $(x =) e + \frac{1}{e}$ do not ISW
 accept values of AWRT 3.08, 3.09
 if M0 then $x = 2e$ with or without working scores SC1

A1

3

(ii) $x \geq 2e$
 accept values of AWRT 5.42, 5.43, 5.44

B1

$e < x \leq e + \frac{1}{e}$
 accept values of AWRT 2.72, 3.08, 3.09
 if B2 not earned, then SC1 for any of
 $e \leq x \leq e + \frac{1}{e}$, $e < x < e + \frac{1}{e}$, $e \leq x < e + \frac{1}{e}$

B2

3

[13]

Q16.

(a) $P = 500e^{\frac{1}{8} \times 60}$
 Must use $t = 60$

EXAM PAPERS PRACTICE M1
 = 904 000
 Nearest thousand required 904000 only

A1

2

(b) (i) $\left(e^{\frac{1}{8}t}\right)^2 = \frac{500000}{500}$

M1

$t = 8 \ln \sqrt{1000}$
 OE Take logs correctly leading to expression for t .

M1

$t = 27.6$ (minutes)
 Accept 27.631

A1

3

Alternative

$$e^{\frac{1}{8}t} = 1000e^{\frac{1}{8}t} \Rightarrow e^{\frac{1}{4}t} = \frac{500000}{500}$$

(M1)

$$t = 4 \ln 1000$$

Take logs correctly leading to expression for t.

(M1)

$$t = 27.6 \text{ (minutes)}$$

(A1)

(3)

Alternative

$$e^{\frac{1}{8}t} = 1000e^{\frac{1}{8}t} \Rightarrow \ln\left(e^{\frac{1}{4}t}\right) = \ln 1000 + \ln\left(e^{\frac{1}{8}t}\right)$$

Take logs correctly.

(M1)

$$t = 4 \ln 1000$$

(M1)

$$t = 27.6 \text{ (minutes)}$$

(A1)

27.6 as final answer NMS 3/3

27.6 following wrong working AO (FIW) but could still score M mark(s)

(3)

$$(ii) \quad 500e^{\frac{1}{8}t} - 500000e^{\frac{1}{8}t} = 45000$$

Set up equation; condone one error; allow in t.

Condone inequality.

M1

$$\times \frac{e^{\frac{1}{8}t}}{500} \Rightarrow \left(e^{\frac{1}{8}t}\right)^2 - 1000 = 90e^{\frac{1}{8}t}$$

$$\left(e^{\frac{1}{8}t}\right)^2 - 90e^{\frac{1}{8}t} - 1000 = 0$$

Multiply by $e^{\frac{1}{8}t}$ and rearrange to AG, be convinced.

A1

$$e^{\frac{1}{8}t} = 100 \quad (e^{\frac{1}{8}t} = -10 \text{ rejected})$$

Solve quadratic equation (retaining positive root).

M1

$$t = 36.8 \text{ (minutes)}$$

CAO

A1

M1 for solve quadratic equation. Let $x = e^{\frac{1}{8}t}$ solve quadratic equation $x^2 - 90x - 1000 = 0$ by inspection, $x = 100$ seen; factors $(x - 100)(x + 10)$ with 100 and 10 seen; complete square $x = 45 \pm \sqrt{3025}$ all correct.

formula $\frac{90 \pm \sqrt{90^2 + 4000}}{2}$ all correct.

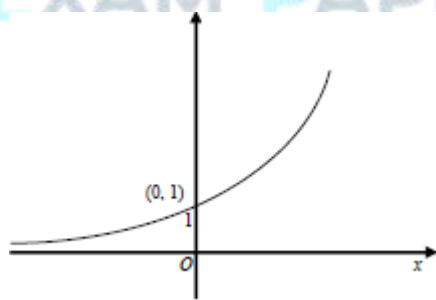
Final answer must have $t = 36.8$ for A1

4

[9]

Q17.

(a)



Correct shape, curve in 1st two quadrants only, crossing positive y-axis once and asymptotic to negative x-axis.

B1

Coordinates (0, 1). Accept y-intercept indicated as 1 on diagram or stated as 'intercept = 1' B0 if graph clearly drawn crossing axes at more than one point

B1

2

- (b) (i) $y^2 - 12 = y$ OE; $7^{2x} - 12 = 7^x$ OE
Eliminates either x or y correctly

M1

$$(y - 4)(y + 3) (= 0); (7^x - 4)(7^x + 3) (= 0)$$

Correct factors or $y = \frac{1 \pm \sqrt{49}}{2}$ or better or $7^x = \frac{1 \pm \sqrt{49}}{2}$
 or better

A1

Since $y (= 7^x) > 0$, [$y (= 7^x) \neq -3$]
 (there is exactly one point of intersection)

*Clear indication that c 's negative solution(s) has / have
 been considered and rejected*

E1

y -coordinate is 4

B1

4

(ii) $7^x = 4$ so $x \log 7 = \log 4$ [or $x = \log_7 4$]

*OE ft on $7^x = k$, where k is positive, to either
 $x \log 7 = \log k$ or $x = \log_7 k$*

M1

$$x = 0.712(414...) = 0.712 \text{ to 3SF}$$

*Condone > three significant figures. If use of logarithms not
 explicitly seen then score 0 / 2*

A1

2

[8]

Q18.

(a) (i) $1000 \times 1.03^5 \approx (\pounds)1160$

Condone missing $\pounds$ sign; 1160 only.

B1

1

(ii) $2000 < 1000 \left(1 + \frac{3}{100}\right)^n$

Condone '=' or '<' used throughout

B1

$$\ln 2 < n \ln 1.03$$

Take logs, any base, of their initial expression **correctly**

M1

$$(n > 23.449\dots) \quad (N =) 24$$

Condone 23

A1

3

$$(b) \quad 1000 \times \left(1 + \frac{3}{100}\right)^n > 1500 \times \left(1 + \frac{1.5}{100}\right)^n$$

Condone use of T for n

Condone '=' or '<' used throughout

B1

$$\ln 1000 + n \ln 1.03 > \ln 1500 + n \ln 1.015$$

Take logs, any base, of their initial expression **correctly**

M1

$$n > \frac{\ln(1.5)}{\ln\left(\frac{1.03}{1.015}\right)}$$

Correct expression for n or T

A1

$$(n > 27.63\dots) \quad (T =) 28$$

Condone 27

A1

4

[8]

Q19.

$$(a) \quad (i) \quad ar^2 = 36; \quad ar^5 = 972;$$

For $ar^2 = 36$ or $ar^5 = 972$ or for seeing $36r^3 = 972$

M1

$$r^3 = \frac{972}{36} (= 27) \Rightarrow r = 3$$

CSO AG Full valid completion.

A1

2

(ii) $a \times 3^2 = 36$
OE. PI

M1

$a = 4$

Correct answer without working scores the two marks

A1

2

(b) (i) $\sum_{n=1}^{20} u_n = S_{20} = \frac{a(1-r^{20})}{1-r}$
OE

M1

$= \frac{4(1-3^{20})}{-2} = -2(1-3^{20}) = 2(3^{20}-1)$

CSO AG Be convinced

A1

2

(ii) $u_n = a \times 3^{n-1}$
Seen or used

B1

$= 4 \times 3^{n-1} > 4 \times 10^{15} \Rightarrow 3^{n-1} > 10^{15}$

$(n-1) \log 3 (>) \log 10^{15}$

Or finds values of u_n for appropriate adjacent integer values of n so that u_n 's are either side of 4×10^{15}

M1

$n-1 > \frac{15}{\log_{10} 3}; n-1 > 31.4...$

($n > 32.4...$ and n is an integer so least value of n is)

$n = 33$

CSO

A1

3

Q20.

(a) $m = 10 \times 2^{-\frac{14}{8}}$

M1

$\approx 3 \text{ (gm)}$

*Condone 2.97 or better
NOT 2.9 as final answer*

A1

2

(b) $2^{-\frac{d}{8}} = \frac{1}{16}$

M1

$\frac{d}{8} = 4 \Rightarrow d = 32$

CSO

A1

2

(c) $0.01m_0 = m_0 \times 2^{-\frac{t}{8}}$

m_0 can be numerical

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M1

$\ln(0.01) = -\frac{t}{8} \ln(2)$

*Take logs correctly from their equation leading
to a linear equation in t .*

M1

$t = 53.15$

$n = 54$

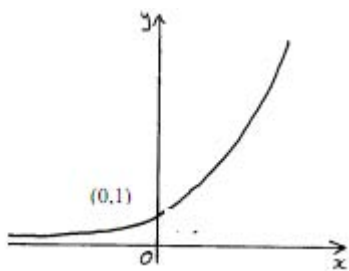
CSO

A1

3

Q21.

(a)



Any graph only crossing the y -axis at $(0, 1)$ stated /indicated ... (accept 1 on y -axis as equivalent) and not drawn **below** x -axis

B1

Correct shaped graph, must clearly go below the intersection pt and an indication of correct behaviour of curve for large positive and large negative values of x . Ignore any scaling on axes.

B1

2

(b) Translation;

Accept 'transl...' as equivalent
[T or Tr is NOT sufficient]

B1

$$\begin{bmatrix} 0 \\ -5 \end{bmatrix}$$

If vector not given, accept **full** equivalent to vector in words provided linked to 'transl../move/shift'
(B0B0 if >1 transformation)

B1

2

(c) (i) $4^x = (2^2)^x = 2^{2x} = (2^x)^2 = Y^2$

$$2^{x+2} = 2^x \times 2^2 = 4Y$$

Justifying either $4^x = Y^2$ or $2^{x+2} = 4Y$ with no errors seen

M1

$$4^x - 2^{x+2} - 5 = 0 \Rightarrow Y^2 - 4Y - 5 = 0$$

AG Be convinced; must have justified both of the above.

A1

2

(ii) $(Y - 5)(Y + 1) = 0$

*Correct factorising or use of quadratic formula
or completing sq. Pl by both solns 5 & -1 seen*

M1

(Since) $2^x > 0$ (for all real x), $2^x = 5$ so only one (real) solution

Rejection of 2^x (condone Y) negative, with justification, (condone " 2^x not negative") followed by statement

E1

$$\log 2^x = \log 5 \Rightarrow x \log 2 = \log 5$$

*Eqn of form $p^x = q \Rightarrow x \log p = \log q$ provided
 $p > 0$ & $q > 0$ OE eg $x = \log_2 5$*

M1

$$x = 2.3219... = 2.322 \text{ (to 3dp)}$$

Condone > 3dp but must see explicit use of logs and must only be the one solution.

A1

4

[10]

Q22.

(a) (i) $(A =) 80$

Ignore units

B1

1

(ii) $2000 = A \times k^{25}$

A or their value from (a)(i)

M1

$$k = \sqrt[25]{25} \text{ or } 25^{\frac{1}{25}}$$

$$\text{or } k = 10^{0.04 \log 25} \text{ or } e^{0.04 \ln 25}$$

$$\Rightarrow k = 1.137411$$

AG

*Correct expression for k ,
or 1.13741146.....seen,*

and correct answer to 6 d.p.

A1

2

(b) $\ln\left(\frac{100000}{\text{their } A}\right) = t \ln k$

Take logs correctly.

Condone miscopied k

$\ln 1250 = t \ln k$ or $t = \log_k 1250$

M1

$t = 55.38$

Condone 55.3 or 55.4 PI

A1

$\Rightarrow 2016$

A1

3

Alternative

By trial and improvement

$1250 = k^t$

Attempt to calculate k^{55} and k^{56} .

(M1)

$t = 56$ or $55 < t < 56$

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(A1)

$\Rightarrow 2016$

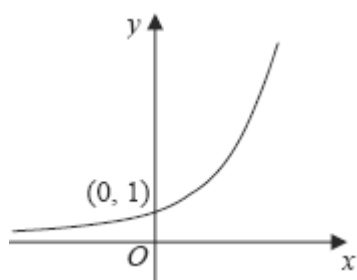
(A1)

(3)

[6]

Q23.

(a)



Shape with some indication of asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis at (0, 1) stated/indicated ... (accept 1 on y-axis as equivalent)

B1

2

- (b) (i) $h = 0.5$
PI

B1

$$\begin{aligned}
 f(x) &= 2^x \\
 I &\approx h/2\{\dots\} \\
 \{\dots\} &= f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)] \\
 &\text{OE summing of areas of the 4 'trapezia'}
 \end{aligned}$$

M1

$$\begin{aligned}
 \{\dots\} &= 1 + 4 + 2(\sqrt{2} + 2 + \sqrt{8}) \\
 &= 5 + 2 \times 6.2426\dots = 17.485\dots \\
 &\text{OE Accept 2 dp (rounded or truncated) as evidence for surds}
 \end{aligned}$$

A1

$$\begin{aligned}
 (I \approx) 4.3713\dots &= 4.37 \text{ (to 3 sf)} \\
 &\text{CAO Must be 4.37} \\
 &\text{SC for those who use 5 strips, max possible is B0M1A1A0}
 \end{aligned}$$

A1

4

- (ii) Increase the number of ordinates
OE

E1

1

(c) Translation;

*Accept 'translat...' as equivalent
[T or Tr is NOT sufficient]*

B1;

$$\begin{bmatrix} -7 \\ 3 \end{bmatrix}$$

*B1 for each component of the vector.
Condone if the equiv 2 vectors are given.
Accept **full** equivalent to vector(s) in
words provided linked to 'translation/
move/shift' and **correct** directions.
(No marks if **different** transformations)*

B1;B1

3

(d) $8 = 2^k + 3 \Rightarrow 2^k = 5$

Correct subst. and an attempted

rearrangement to $2^k = N$. PI by $k = \frac{\log 5}{\log 2}$

M1

$$k = \log_2 5$$

Accept $m = 2, n = 5$

A1

2

EXAM PAPERS PRACTICE

[12]

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Q24.

(a) (i) $t = 0 \quad h = A(1 - 1) = 0$

B1

1

(ii) $57 = A \left(1 - e^{-\frac{12}{4}} \right)$

M1

$$A = \frac{57}{(1 - e^{-1})} \approx 60$$

Or 59.9... seen.

$A = \text{correct expression} \approx 60 \text{ 2 sf}$

A1

2

(b) (i) $h = 48 \quad \frac{48}{60} = 1 - e^{-\frac{1}{4}t}$

M1

$$\ln\left(e^{-\frac{1}{4}t}\right) = \ln\left(\frac{1}{5}\right)$$

m1

$$-\frac{1}{4}t = -\ln 5 \Rightarrow t = 4 \ln 5$$

A1

3

(ii) $\frac{dh}{dt} = -\frac{1}{4} \times -60 \times e^{-\frac{1}{4}t}$
Differentiate, condone sign errors

M1

$$60e^{-\frac{1}{4}t} = 60 - h \Rightarrow \frac{dh}{dt} = \frac{1}{4}(60 - h)$$

Eliminate $e^{-\frac{1}{4}t}$

m1

$$\frac{dh}{dt} = 15 - \frac{h}{4}$$

CSO, AG

A1

3

(iii) $h = 8$

B1

1

[10]

Q25.

(a) $(y =) 1$

B1

1

(b) $h = 0.2$

PI

B1

$$f(x) = 2^{4x}$$

$$I \approx h/2 \{ \dots \}$$

$$\{ \cdot \} = f(0) + f(1) + 2[f(0.2) + f(0.4) + f(0.6) + f(0.8)]$$

OE summing of areas of the 'trapezia'..

M1

$$\{ \cdot \} = 1 + 16 + 2(2^{0.8} + 2^{1.6} + 2^{2.4} + 2^{3.2})$$

$$= 1 + 16 + 2(1.741.. + 3.031.. + 5.278.. + 9.1895..)$$

$$= [17 + 2 \times 19.24...]$$

OE Accept 2dp rounded or truncated evidence

A1

$$I = 5.55 \text{ (to 2 dp)}$$

Must be 5.55

A1

4

(c) Stretch(I) in y -direction(II) scale factor $\frac{1}{8}$ (III)
*Need (I) **and** either (II) or (III)*

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M1

Need (I) and (II) and (III)

A1

ALTn: Translation with an indication
 that the translation is in the x -direction (B1)

$$\begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$$

(B1)

*Combination of **different**
 transformations scores 0/2*

2

(d) $g(x) = 2^{4(x-1)} - \frac{1}{2}$

$$B1 \text{ for either } 2^{4(x+1)} - \frac{1}{2} \text{ or for } 2^{4(x-1)} + \frac{1}{2} \text{ or for } 2^{4x-1} - \frac{1}{2}$$

B2,1,0

$$\text{At } Q, y = 0 \Rightarrow 2^{4(x-1)} = 2^{-1}$$

Reaches a stage from which linear eqn can be stated directly eg an alternative stage is

$$4(x-1) \log 2 = -\log 2$$

M1

$$\Rightarrow 4x - 4 = -1 \Rightarrow x = 0.75$$

NMS mark as 4 or 0

A1

4

(e) (i) $\log_a k = \log_a 2^3 + \log_a 5 - \log_a 4$

$$\log_a k = \log_a (2^3 \times 5) - \log_a 4$$

One law of logs used

M1

A second law of logs used; could be

$$\log_a k = \log_a 2^3 + \log_a \left(\frac{5}{4}\right)$$

M1

$$\log_a k = \log_a \left(\frac{2^3 \times 5}{4}\right) = \log_a 10 \Rightarrow k = 10$$

CSO AG

A1

3

(ii) $2^{4x-3} = \frac{5}{4}$ so

Equate y's, take logs (to any base) of both sides and apply 3rd law of logs.

$$(4x-3) \log_{10} 2 = \log_{10} \frac{5}{4}$$

M1

$$x = \frac{3 \log_{10} 2 + \log_{10} \left(\frac{5}{4} \right)}{4 \log_{10} 2}$$

m1

$$x = \frac{\log_{10} 10}{4 \log_{10} 2} \text{ so } x = \frac{1}{4 \log_{10} 2}$$

A1

Altn $4x \log 2 = \log \left(\frac{5}{4} \times 2^3 \right)$

Rearrange correctly to $x = \dots$

Altn $4x \log 2 = \log 10$

In both cases, log term(s) must have same base and expressions must be in an exact form, ie not approx. dec. vals

CSO AG Must be clear evidence that base 10 is used, also be convinced

3

[17]

Q26.

(a) $y = e^x \rightarrow e^{2x} - 1$

Stretch (I)

scale factor $\frac{1}{2}$ (II)

I + (II or III)

M1

in x -direction (III)

I + II + III

A1

Translation

Allow "translate"

E1

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

OE "1 unit down" etc

B1

4

(b) $x = 0 \quad y = 6$ or $(0, 6)$

Both coordinates must be stated, not simply 6 marked on diagram

B1

1

(c) (i) $e^{2x} - 1 = 4e^{-2x} + 2$
 $e^{4x} - e^{2x} = 4 + 2e^{2x}$
 or $(e^{2x})^2 - e^{2x} = 4 + 2e^{2x}$

Multiplying both sides by e^{2x}

M1

$$(e^{2x})^2 - 3e^{2x} - 4 = 0$$

AG With no errors seen

A1

2

(ii) $(e^{2x} - 4)(e^{2x} + 1)$
 $(e^{2x} \pm 4)(e^{2x} \pm 1)$

M1

$$x = \ln 2 \text{ or } \frac{1}{2} \ln 4$$

A1

Reject $e^{2x} = -1$ OE

Eg $e^{2x} > 0$, $e^{2x} \neq -1$, impossible etc

A1

3

(d) $\int (4e^{-2x} + 2) dx$ (I)

$$= \left[\frac{4e^{-2x}}{-2} + 2x \right]_0^{\ln 2}$$

I or II attempted and e^{-2x} or e^{2x} integrated correctly

M1

$$= \left(\frac{4e^{-2\ln 2}}{-2} + 2\ln 2 \right) - \left(\frac{4}{-2} + 0 \right)$$

F ['their $\ln 2$ ' from (c)(ii)] - F[0]

m1

$$= -\frac{1}{2} + 2 \ln 2 + +2 = \frac{3}{2} + 2 \ln 2$$

$$\int (e^{2x} - 1) \, dx \quad (\text{II})$$

$$= \left[\frac{e^{2x}}{2} - x \right]_0^{\ln 2}$$

Both I and II correctly integrated

A1

$$= \left(\frac{e^{2 \ln 2}}{2} - \ln 2 \right) - \left(\frac{1}{2} - 0 \right)$$

$$= 2 - \ln 2 - \frac{1}{2} = \frac{3}{2} - \ln 2$$

$$A = \left(\frac{3}{2} + 2 \ln 2 \right) - \left(\frac{3}{2} - \ln 2 \right)$$

Attempt to find difference of 'their I – their II'

B1ft

$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \quad \text{OE}$$

CSO must be exact

A1

Alternative

$$A = \int (4e^{-2x} + 2) \, dx - \int (e^{2x} - 1) \, dx$$

Condone functions reversed

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$$= \int_{(0)}^{(\ln 2)} (4e^{-2x} - e^{2x} + 3) \, dx$$

$$= \left[\frac{4e^{-2x}}{-2} - \frac{e^{2x}}{2} + 3x \right]_0^{\ln 2}$$

e^{2x} or e^{-2x} correctly integrated

(M1)(A1)

$$= \left(-2e^{-2 \ln 2} - \frac{1}{2}e^{2 \ln 2} + 3 \ln 2 \right) - \left(-2 - \frac{1}{2} \right)$$

Correct substitution of their $\ln 2$ from (c)(ii) into their integrated expression

(m1)

$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \text{ OE}$$

CSO must be exact

(A1)

5

[15]

Q27.

(a) (i) $\log_a 40$

Accept 'k = 40'

B1

1

(ii) $\log_a 8$

Accept 'k = 8'

B1

1

(iii) $\log_a 125$

Accept 'k = 125' but not 'k = 5³'

B1

1

(b) $\log_{10} \{(1.5)^{3x}\} = \log_{10} 7.5$

Correct statement having taken logs of both sides of $(1.5)^{3x} = 7.5$ OE PI or $3x = \log_{1.5} 7.5$ seen

M1

$$3x \log_{10} 1.5 = \log_{10} 7.5$$

$$\log 1.5^{3x} = 3x \log 1.5 \text{ OE}$$

m1

$$x = \frac{\lg 7.5}{3 \lg 1.5} = 1.65645 \dots = 1.656 \text{ to 3 dp}$$

Both method marks must have been awarded with clear use of logarithms seen

A1

3

(c) $\log_2 p = m \Rightarrow p = 2^m; \log_8 q = n \Rightarrow q = 8^n$

Either $p = 2^m$ or $q = 8^n$ seen or used

M1

$$p = 2^m \text{ and } q = 2^{3n}$$

Writing $8^n = 2^{3n}$ and having $p = 2^m$

m1

$$pq = 2^m \times (2^3)^n = 2^m \times 2^{3n} \text{ so } pq = 2^{m+3n}$$

Accept $y = m + 3n$

A1

3

[9]

Q28.

(a) $3e^x = 4$

$$e^x = \frac{4}{3}$$

M1

$$x = \ln \frac{4}{3}$$

A1

2

(b) (i) $3e^x + 20e^{-x} = 19$

$$3y + \frac{20}{y} = 19 \text{ or } 3e^{2x} + 20 = 19e^x$$

$$3y^2 - 19y + 20 = 0$$

AG

B1

1

(ii) $(3y - 4)(y - 5) = 0$

$$y = \frac{4}{3}, 5$$

B1

$$\therefore x = \ln \frac{4}{3}, \ln 5$$

In (their + ve y 's)

M1A1

3

[6]

Q29.

(a) (i) $\sqrt{125} = \sqrt{25 \times 5} = 5\sqrt{5}$

OE eg $\sqrt[3]{125} = \sqrt[3]{5^3}$ or $5^{1.5}$ seen

M1

$$5^p = \sqrt{125} \Rightarrow p = 1.5$$

Correct value of p must be explicitly stated

A1

Alternative:

$$p \log 5 = \frac{1}{2} \log 125$$

2

OE eg $p \log 5 = \log 11.18$

or eg $p = \log_5 \sqrt{125}$

(M1)

$$p \log 5 = \frac{3}{2} \log 5 \Rightarrow p = \frac{3}{2}$$

Correct value of p must be explicitly stated

(A1)

(ii) $5^{2x} = \sqrt{125} = 5^p \Rightarrow x = 0.5p = 0.75$

Must be $0.5 \times c$'s value of p

SC: $x = 0.75$ with c's ans (a)(i) $5^{1.5}$ scores B1F

B1F

1

(b) $3^{2x-1} = 0.05$

$$(2x - 1) \log 3 = \log 0.05$$

Take logs of both sides and use 3rd law of logs. Pl eg by $2x - 1 = \log_3 0.05$ seen

M1

$$x = \frac{\log_{10} 0.05}{2 \log_{10} 3} + \frac{1}{2}$$

Correct rearrangement to $x = \dots PI$

m1

$$= -0.8634(165\dots) = -0.8634 \text{ to 4dp}$$

*Condone > 4 dp. Must see logs clearly
used in solution, so NMS scores 0/3*

A1

3

(c) $\log_a x = 2(\log_a 3 + \log_a 2) - 1$
 $= 2 \log_a (3 \times 2) - 1$

A valid law of logs used

M1

$$= \log_a (6^2) - 1$$

Another valid law of logs used

M1

$$= \log_a 36 - \log_a a$$

$\log_a a = 1$ quoted or used

or $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$

OE

B1

$$\log_a x = \log_a \left(\frac{36}{a} \right) \Rightarrow x = \frac{36}{a}$$

$$x = \frac{36}{a} \text{ or } x = 36a^{-1}$$

CSO Must be

A1

4

[10]

Q30.

(a) (i) $A = 12499$

Stated in (i) or (ii)

B1

1

(ii) $k^{36} = \frac{7000}{\text{their } A}$

$$p = \frac{7000}{12499} = 0.560044803$$

M1

$$k = \sqrt[36]{0.56(00448\dots)} = 0.9840251(26)$$

$$\text{or } (0.56(00448\dots))^{\frac{1}{36}}$$

$$\text{or } k = \sqrt[36]{\frac{7000}{12499}}$$

$$k = 0.984025$$

Correct expression for k or 7th dp seen.

$$k = 10^{\frac{1}{36} \log p} \quad \text{or } k = 10^{-0.00699\dots}$$

$$k = e^{\frac{1}{36} \ln p} \quad \text{or } k = e^{-0.016103\dots}$$

AG

A1

2

(b) $k^t = \frac{5000}{\text{their } A}$

$$\frac{5000}{12499} = 0.400032\dots; \text{condone } 4999$$

M1

$$t \log(k) = \log\left(\frac{5000}{A}\right) \quad (t = 56.89)$$

Correct use of logs

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$$n = 57$$

$$n \text{ integer}; \quad n = 57 \quad \text{CAO}$$

A1

Alternative ; trial and improvement on

$$5000 = 12499 \times 0.984025'$$

2 values of $t \geq 40$

(M1)

$$1 \text{ value of } t \quad 50 < t < 60$$

(m1)

$$n = 57$$

(A1)

Special case, answer only

$$n = 57 \quad 3/3$$

$$n = 56 \quad 0/3$$

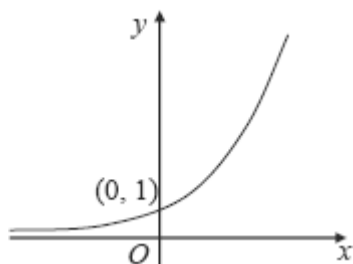
$$n = 56.9 \quad 2/3$$

3

[6]

Q31.

(a)



Shape (graph must clearly go below the intersection pt.). Condone if x -axis is a tangent

B1

Only intersection with y -axis at $(0, 1)$ stated/indicated ... (accept 1 on y -axis as equivalent) 0

B1

2

- (b) (i) Stretch (I) in x -direction (II) scale factor 0.5 (III)

Need (I) & one of (II), (III)

M1

M0 if >1 transformation

A1

2

- (ii) Translation;

Must be 'Translation' or 'translate (d)' for 1st B mark

B1;

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

*Accept **full** equivalent to vector in words provided linked to 'translation'*

*move/shift' and **negative** x-direction
(Note: B0 B1 is possible)*

B1

ALTn: Stretch (I) in y-direction (II) scale factor 3 (III)
[Mark the alternative as in (b)(i).]

2

(c) (i) $9^x = (3^2)^x = 3^{2x} = (3^x)^2 = Y^2$;
Justifying either $9^x = Y^2$ or $3^{x+1} = 3Y$

M1

$$\begin{aligned} 3^{x+1} &= 3^x \times 3^1 = 3Y \\ 9^x - 3^{x+1} + 2 &= 0 \Rightarrow Y^2 - 3Y + 2 = 0 \\ \Rightarrow (Y-1)(Y-2) &= 0 \\ \text{AG} \end{aligned}$$

A1

2

(ii) $Y = 1 \Rightarrow 3^x = 1 \Rightarrow x = 0$
AG (Accept direct substitution if convinced)

B1

$$\begin{aligned} Y = 2 &\Rightarrow 3^x = 2 \\ \log_{10} 3^x &= \log_{10} 2 \\ \text{Takes logs of both, PI by 'correct' value(s) later.} \\ \text{or } x &= \log_3 2 \text{ seen} \end{aligned}$$

M1

$$x \log_{10} 3 = \log_{10} 2$$

Use of $\log 3^x = x \log 3$ or

$$\log_3 2 = \frac{\lg 2}{\lg 3} \quad \text{OE (PI by } \log_3 2 = 0.630 \text{ or } 0.631 \text{ or better)}$$

m1

$$x = \frac{\lg 2}{\lg 3} = 0.630929 \dots = 0.6309 \text{ to 4 dp}$$

Must show that logarithms have been used
otherwise 0/3

A1

4

Q32.

(a) (i) $A = 20$

B1

1

(ii) $\frac{2000}{A} = k^{60}$

M1

$$k = (100)^{\frac{1}{60}} = 1.079775$$

AG, or $k = 10^{\frac{\log 100}{60}} = 10^{0.0333}$ or $\sqrt[60]{100}$ or
 $\sqrt[30]{10}$ or $e^{\frac{\ln 100}{60}} = e^{0.076}$ or $e^{0.077}$ or
 1.0797751(6) seen

A1

2

(iii) $P = 20 \times k^{2008-1885}$

M1

$= 251780 \approx 252000$
 CAO nearest 1000

A1

2

(b) $15 \times 1.082709 = 20 \times 1.079775^t$
 equate prices

M1

$$\frac{15}{20} = \left(\frac{1.079775}{1.082709} \right)^t$$

*t as a single index, or correct log expression
 at this stage*

M1

$$t = \frac{\log 0.75}{\log 0.997290}$$

expression for t

m1

$$t = 106.017 \Rightarrow 1991$$

SC Answer only/Trial and error
 106 seen (2 out of 4)
 1991 (4 out of 4)

A1

4

[9]

Q33.

- (a) (i) $h = 0.5$
 Integral = $\frac{h}{2} \{ \dots \}$
PI

B1

$$\{ \dots \} = f(0) + 2\left[f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right)\right] + f(2)$$

OE summing of areas of the four traps.

M1

$$\begin{aligned} \{ \} &= 1 + 2[\sqrt{6} + 6 + 6\sqrt{6}] + 36 \\ &= 1 + 2[2.449.. + 6 + 14.6969..] + 36 \\ &\text{Condone 1 numerical slip. Accept 3sf} \\ &\text{values if not exact.} \end{aligned}$$

A1

$$\begin{aligned} &= 37 + 2 \times 23.146.. = 83.292... \\ \text{Integral} &= 0.25 \times 83.292.. = 20.8 \text{ (3sf)} \\ &\text{CAO; must be 20.8} \end{aligned}$$

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph
Accept single trapezium with its sloping side above the curve

M1

{Approximation is an} overestimate
Dep. on 4 trapezia with each of their upper vertices lying on the curve

A1

2

- (b) (i) Stretch (I) in x -direction (II)
Need (I) and one of (II), (III)
M0 if more than one transformation

M1

(scale factor) $\frac{1}{3}$ (III)

A1

2

(ii) $6^{3x} = 84$
PI

M1

$$\log_{10} 6^{3x} = \log_{10} 84$$

*Take logs of both sides of $a^x = b$, PI by
 'correct' value(s) later or $3x = \log_6 84$*

M1

$$3x \log_{10} 6 = \log_{10} 84$$

*Use of $3 \log 6^{3x} = 3x \log 6$ OE
 or $3x = \log_6 84$ seen*

m1

$$x = \frac{\lg 84}{3 \lg 6}$$

$$x = 0.82429\dots = 0.824 \text{ (to 3dp)}$$

*Must see that logs have been used before
 any of the last 3 marks are awarded in
 (b)(ii). Condone > 3dp*

A1

4

(c) $f(x) = 6^{x-1} - 2$
B1 For either $6^{x-1} + 2$ or for $6^{x+1} - 2$

B2,1

2

[14]

Q34.

- (a) (i) $p \frac{dx}{dt} = q$
 where p and q are functions

M1

$$\frac{dx}{dt} = -kx$$

in any correct combination

A1

2

- (ii) $-500 = -k \ 20000$ or $500 = k \ 20000$
 condone sign error or missing 0
 k can be on either side of the equation

M1

$$k = \frac{5}{500} \quad (= 0.025)$$

CSO both (a)(i) and (a)(ii)

A1

2

- (b) (i) $A = 1300$

B1

1

(ii) $100 > Ae^{-0.05t}$

condone = for >; condone 99 for 100

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M1

$$\ln \left(\frac{100}{A} \right) > -0.05t$$

take logs correctly
 condone 0.5

m1

$$t > 51.3$$

or by trial and improvement
 (see end of mark scheme)

A1

population first exceeds 1900 in 2059

F if M1 m1 earned and $t > 0$ following A

A1F

4

ALTERNATIVE METHOD

$$t = 51 \rightarrow 101.5$$

$t = 51$ or $t = 52$ considered

M1

$$t = 52 \rightarrow 96.6$$

$$\Rightarrow 51 < t < 52$$

population first exceeds 1900 in 2059

CAO

A3

4

[9]

Q35.

(a) (i)

$$\int (4 - e^{2x}) dx$$

$$= 4x - \frac{1}{2} e^{2x} (+c)$$

4x

B1

$$-\frac{1}{2} e^{2x}$$

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B1

2

$$(ii) \int_0^{\ln 2} = \left[4x - \frac{1}{2} e^{2x} \right]_0^{\ln 2}$$

$$= \left[4 \ln 2 - \frac{1}{2} e^{2 \ln 2} \right] - \left[(0) - \frac{1}{2} (e^0) \right]$$

Substitute both $\ln 2$ and 0 correctly into an integrated expression

M1

$$= 4 \ln 2 - 2 + \frac{1}{2}$$

Convincing

$$= 4 \ln 2 - \frac{3}{2}$$

AG

A1

2

(b) (i) $x = 0$

$$y = 4 - 1 = \underline{\underline{3}}$$

B1

1

(ii) At B , $y = 0$

$$4 - e^{2x} = 0$$

Or reverse argument

M1

$$e^{2x} = 4$$

$$x = \ln 2$$

AG

A1

2

(c) $\frac{dy}{dx} = -2e^{2x}$

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B1

$x = \ln 2$, Gradient $= -2e^{2 \ln 2} = -8$

$x = \ln 2$ into ke^{2x}

M1

$$\text{Gradient normal} = \frac{1}{8} = \frac{1}{2e^{2 \ln 2}}$$

OE

A1

Equation $y = -\frac{1}{8}x - \frac{1}{8} \ln 2$

OE

A1

4

(d) When $x = 0$

$$y = -\frac{1}{8} \ln 2$$

*Attempt to integrate their line and substitute
 $x = 0, \ln 2$*

M1

Area $\Delta = \frac{1}{16} (\ln 2)^2$ condone – ve sign = 0.03

$$\frac{1}{2} (\text{their } y) \times \ln 2$$

A1 ✓

Total area $4 \ln 2 - \frac{3}{2} + \frac{1}{16} (\ln 2)^2 = 1.30$

AWRT

CSO

A1

3

[14]



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